Sample Student Portfolio

Scientific Writing and Research Communication in Physics

Worked model for the seven practical items — B.Sc. (Honours) Physics, Semester VI

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1. Portfolio Overview

This portfolio presents a worked model for the Skill Enhancement Course (SEC) *Scientific Writing and Research Communication in Physics* (course code MG6SECPHY302), offered in Semester VI of the B.Sc. (Honours) Physics programme. Gravitational waves are used as a common scientific theme to demonstrate how physics research can be identified, documented, analysed, written about, and communicated to different audiences.

The portfolio brings together seven practical items covering reference management, visual communication, critical reading, and scientific writing. The activities progress from building a reference database from reliable sources to analysing research publications, review papers, scientific posters, doctoral theses, and popular science articles. Together, they illustrate that effective scientific communication requires more than presenting correct information: it also involves evaluating evidence, organising ideas, acknowledging sources, interpreting data responsibly, and adapting the presentation to the intended audience.

Submission Requirements

The completed portfolio should include:

- a cover page containing the student's name, register number, course details, and a brief statement identifying the chosen theme;
- a contents page listing all portfolio items;
- all seven practical submissions;
- appropriate citations and a consistent reference list and
- a signed declaration confirming that the submitted work is the student's own and that all sources have been acknowledged.

Portfolio Components

Unit	Item	Expected submission	Length	Marks*
3.1	Bibliography database	An exported reference library containing at least 20 entries, a formatted bibliography in a consistent citation style, and a short note explaining how the references were checked.	2–3 pages	3
3.2	Research article analysis	A critical analysis of one open-access research article, using the prescribed analysis framework.	600–800 words	3
3.3	Poster portfolio	Two scientific posters in PDF format, each accompanied by a short design note. The portfolio should incorporate the data table and graph prepared for Unit 2.2.	2 posters	4
3.4	Review paper analysis	An analysis of a topical review paper, including a concept map showing the relationships among its principal ideas.	600–800 words	3
3.5	Poster/slides analysis	A critical analysis of an existing scientific poster or slide presentation, accompanied by a redesign sketch.	500–700 words	2
3.6	Thesis analysis	An analysis of an open-access thesis, focusing on its organisation, academic writing, use of evidence, and referencing.	600–800 words	2
3.7	Popular science appreciation	An appreciation of a popular science article or book chapter, considering its scientific accuracy, style, accessibility, and effectiveness in communicating science.	500–700 words	3
	Total suggested allocation			**20**

**Assessment note:* The syllabus specifies 20 marks for the portfolio/record (within the 75 marks of the semester-end practical evaluation) but does not prescribe a mark distribution for individual items. The allocation above is therefore a suggested model and should be adjusted to the agreed assessment practice.

How to Use This Model

This document is intended as a worked example to guide students in preparing their own portfolios. Gravitational waves provide a common theme for the examples, but students should follow the course requirements and any instructions issued by the teacher when

selecting sources and preparing their submissions. The examples should be studied critically rather than copied. Students are expected to make their own choices, verify bibliographic details and scientific claims, write in their own words, and explain the decisions made during each activity.

The portfolio should demonstrate not only the completion of the prescribed tasks but also the development of the student's ability to read scientific literature critically, communicate physics clearly, and practise responsible academic writing.

2. Portfolio Item : Bibliography database

Introduction: Built a library of journal articles / books and other publications in Zotero (or JabRef), imported entries by DOI, corrected the metadata by hand, tagged them (for example `detector`, `review`, `open-data`), and exported to BibTeX. A formatted list in APS style is given. The sources were found with the search methods of Unit 2.1; a list of the journals and data sources consulted is given in the optional extension at the end of the portfolio.

BibTeX export (excerpt of two of the entries)

```
@article{Abbott2016GW150914,
  author  = {Abbott, B. P. and others},
  title   = {Observation of Gravitational Waves from a Binary Black Hole Merger},
  journal = {Physical Review Letters},
  volume  = {116}, number = {6}, pages = {061102}, year = {2016},
  doi     = {10.1103/PhysRevLett.116.061102}
}
@article{Sathyaprakash2009LRR,
  author  = {Sathyaprakash, B. S. and Schutz, Bernard F.},
  title   = {Physics, Astrophysics and Cosmology with Gravitational Waves},
  journal = {Living Reviews in Relativity},
  volume  = {12}, pages = {2}, year = {2009},
  doi     = {10.12942/lrr-2009-2}
}
```

Formatted list (APS-style)

[1] *Research articles.* B. P. Abbott *et al.* (LIGO Scientific Collaboration and Virgo Collaboration), "Observation of gravitational waves from a binary black hole merger," Phys. Rev. Lett. **116**, 061102 (2016). doi:10.1103/PhysRevLett.116.061102.

[2] B. P. Abbott *et al.*, "GW170817: Observation of gravitational waves from a binary neutron star inspiral," Phys. Rev. Lett. **119**, 161101 (2017). doi:10.1103/PhysRevLett.119.161101.

[3] R. A. Hulse and J. H. Taylor, "Discovery of a pulsar in a binary system," Astrophys. J. **195**, L51 (1975). doi:10.1086/181708.

[4] *Instrument and data papers.* J. Aasi *et al.* (LIGO Scientific Collaboration), "Advanced LIGO," Class. Quantum Grav. **32**, 074001 (2015). doi:10.1088/0264-9381/32/7/074001.

[5] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), "Open data from the third observing run of LIGO, Virgo, KAGRA and GEO," Astrophys. J. Suppl. Ser. **267**, 29 (2023). doi:10.3847/1538-4365/acdc9f.

[6] *Reviews.* B. S. Sathyaprakash and B. F. Schutz, "Physics, astrophysics and cosmology with gravitational waves," Living Rev. Relativ. **12**, 2 (2009). doi:10.12942/lrr-2009-2; preprint arXiv:0903.0338.

[7] E. Berti, "Ten years of gravitational-wave astronomy," preprint arXiv:2509.10395 (2025).

[8] *Viewpoint and popular articles.* E. Berti, "Viewpoint: The first sounds of merging black holes," Physics **9**, 17 (2016). doi:10.1103/Physics.9.17; preprint arXiv:1602.04476.

[9] APS News, "September 2015: Physicists detect gravitational waves for the first time" (American Physical Society, 2025). https://www.aps.org/apsnews/2025/08/

detect-gravitational-waves-first-time

[10] The Nobel Prize in Physics 2017, summary page, NobelPrize.org. https://www.nobelprize.org/prizes/physics/2017/summary/

[11] *Thesis.* S. Mitra, "Gravitational waves from inspiraling binaries and cosmological ramifications," Ph.D. thesis, Savitribai Phule Pune University (IUCAA), 2006. hdl:10603/133177 (Shodhganga).

[12] *Poster.* M. Beroiz *et al.*, "Distributing public data with the Gravitational Wave Open Science Center," poster P302, ADASS 2023. https://pretalx.com/adass2023/talk/SVH7HL

Summary / Conclusions

Zotero imported most records correctly from their DOIs, but the thesis record needed its Shodhganga handle added by hand. I then compared every entry against its DOI or repository page, and rebuilt the list in a second style (IEEE) to confirm that the reference manager, not manual typing, controlled the formatting.

3. Portfolio Item : Data presentation and posters

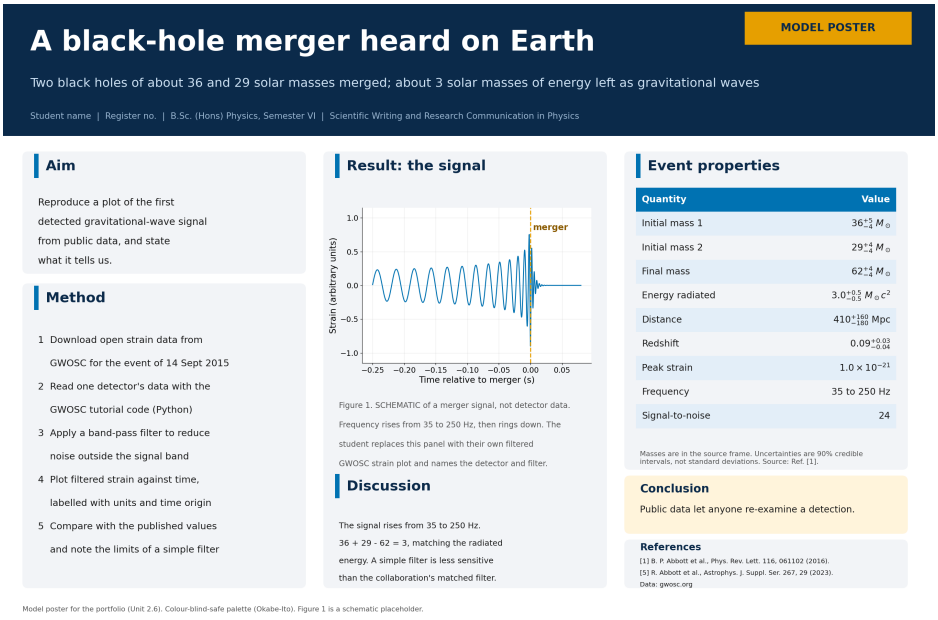
3.1 Data table prepared for the posters

All values below come from the abstract of the first-detection paper (Abbott *et al.* 2016, Ref. [1]). Uncertainties are 90% credible intervals, as the paper states, and they are not standard deviations; the table's caption must say so.

Table 2: Properties of the first gravitational-wave event, GW150914. Source: Ref. [1].

Quantity	Value	Unit
Initial black hole mass 1 (source frame)	36^{+5}_{-4}	$M_\odot$
Initial black hole mass 2 (source frame)	29^{+4}_{-4}	$M_\odot$
Final black hole mass (source frame)	62^{+4}_{-4}	$M_\odot$
Energy radiated in gravitational waves	$3.0^{+0.5}_{-0.5}$	$M_\odot c^2$
Luminosity distance	410^{+160}_{-180}	Mpc
Redshift	$0.09^{+0.03}_{-0.04}$	(dimensionless)
Peak strain	1.0×10^{-21}	(dimensionless)
Signal frequency range	35 to 250	Hz
Matched-filter signal-to-noise ratio	24	(dimensionless)

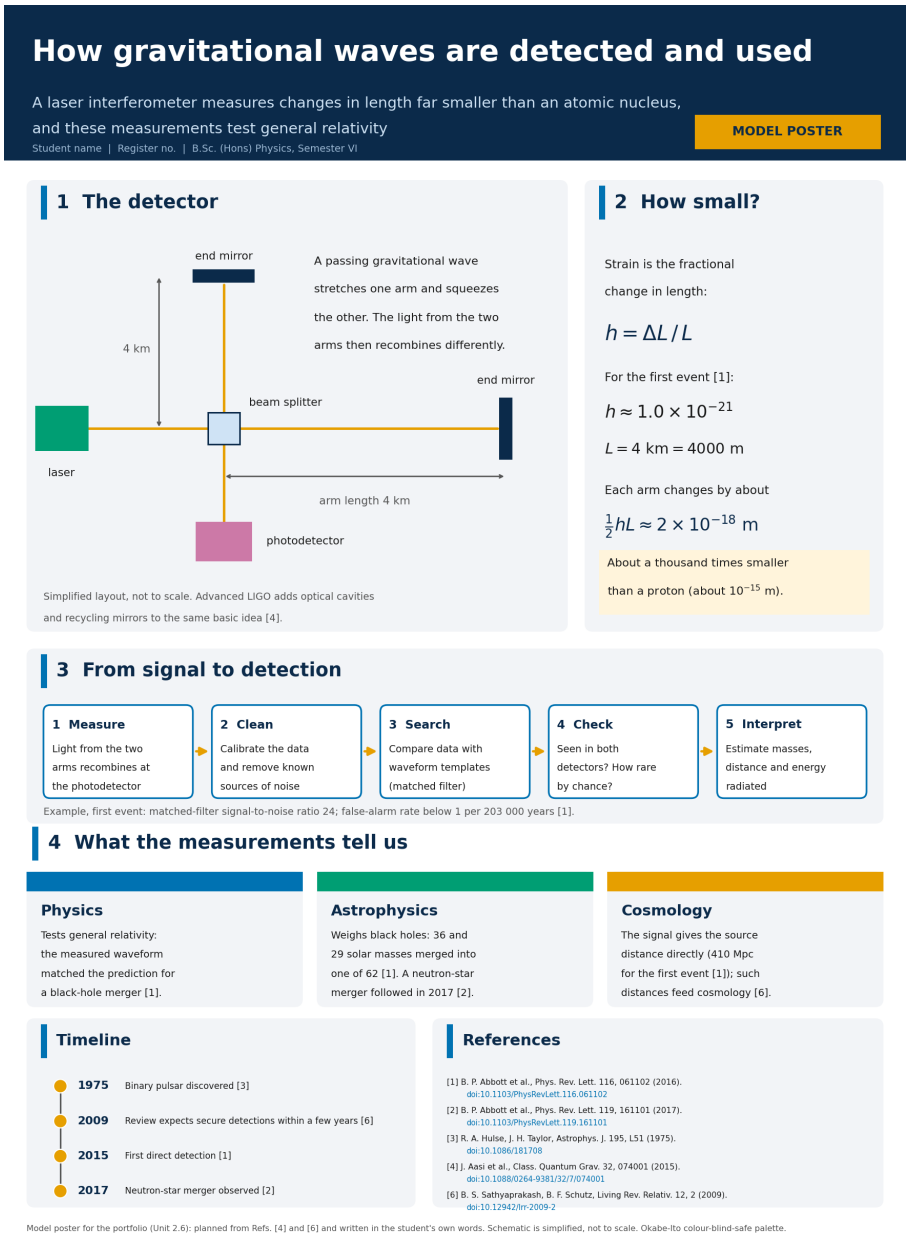
3.2 Poster A: "A black-hole merger heard on Earth"



Poster element	Content planned
Title (the message)	Two black holes of about 36 and 29 solar masses merged, and about 3 solar masses of energy left as gravitational waves
Aim	Reproduce a plot of the first detected signal from public data and summarise what it tells us
Method	Open data from GWOSC; band-pass filtering; plotting in Python using the GWOSC tutorials
Results	Strain plot (large, labelled) and the data table above, with the 90% interval note
Discussion	The signal rises from 35 to 250 Hz; the check $36 + 29 - 62 \approx 3$; limits of a simple filter compared with the collaboration's matched-filter analysis
Conclusion	One sentence only
References	Refs. [1], [5] and the GWOSC site, in the chosen style
Design note	Two-minute test passed by a classmate; figure and table readable at 1 m; four to five short text blocks

3.3 Poster B: "How gravitational waves are detected and used"

Poster element	Content planned
Title (the message)	A laser interferometer measures changes in length far smaller than an atomic nucleus, and these measurements test general relativity
Content source	The review in Ref. [6] (detectors, sources, data analysis and what we learn) and the instrument paper in Ref. [4], summarised, not copied
Figures	A redrawn schematic of an interferometer made by the student; a flow chart "signal → filter → detection"
Design note	Different layout from Poster A; consistent colours; colour-blind-safe palette; references with DOIs



Summary

Made the posters with free tools such as LibreOffice Impress / Inkscape / a LaTeX poster class on Overleaf, and submit both as PDF with a one-paragraph design note.

4. Portfolio Item : Analysis of a research article

Research article analysed. B. P. Abbott *et al.*, "Observation of gravitational waves from a binary black hole merger," Phys. Rev. Lett. **116**, 061102 (2016), open access: https://doi.org/10.1103/PhysRevLett.116.061102.

Frame element	**Model analysis**
Audience and purpose	Written for professional physicists and astronomers, to announce the first direct detection of gravitational waves and the first observation of a binary black hole merger. The journal is a letters journal, so the paper must establish the claim concisely.
Message	On 14 September 2015 two detectors simultaneously saw a transient signal whose frequency rose from 35 to 250 Hz and which matches the waveform predicted by general relativity for a black-hole merger and ringdown.
Structure	Abstract and the different parts of the article *[Student lists the actual section headings from the paper and states what is each part about.]*
Evidence	Significance statement in the abstract: signal-to-noise ratio of 24; false alarm rate below one per 203 000 years, a significance above 5.1σ; source properties with 90% credible intervals.
Visuals	*[Student comments on the figure showing the signals from the two sites: labelling, axes, units, whether the figure stands alone with its caption.]*
Language	Precise, quantitative, with uncertainties reported in a consistent notation; the abstract contains no unexplained jargon beyond what a physicist expects.
Sources and authorship	The author list runs to about a thousand names, reflecting collaboration authorship in large experiments.
Overall	Strength: the abstract states the claim, the evidence and the quantities in a few sentences. Weakness for a general reader: the technical detail needs prior knowledge.

Summary

Abbott *et al.* report the first direct detection of gravitational waves: a signal seen by both LIGO detectors on 14 September 2015, rising from 35 to 250 Hz and matching general relativity's prediction for a binary black hole merger. The evidence is strong (signal-to-noise ratio 24, false-alarm rate below one per 203 000 years), and the source properties are given with 90% credible intervals. The abstract states the claim, the evidence and the key numbers clearly, but the technical detail assumes a physics background, so a general reader is better served by a popular account.

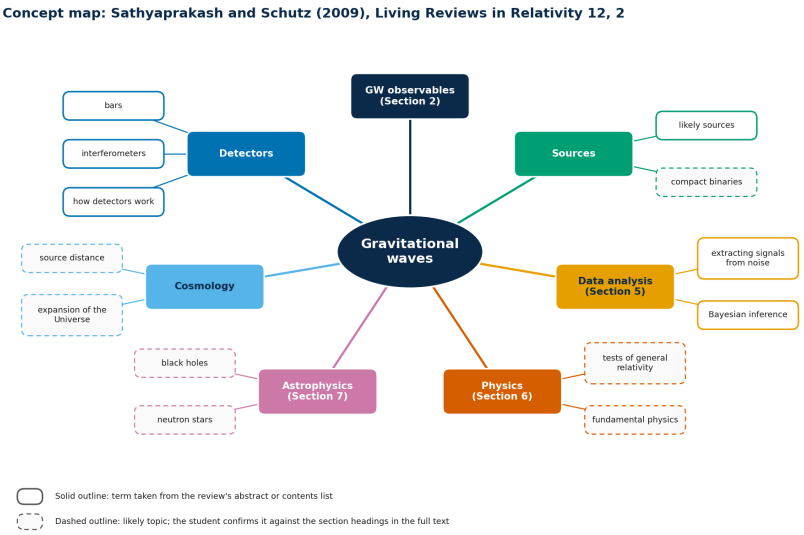
5. Portfolio Item : Analysis of a topical review paper

Review paper analysed. B. S. Sathyaprakash and B. F. Schutz, "Physics, astrophysics and cosmology with gravitational waves," *Living Reviews in Relativity* **12**, 2 (2009). https://doi.org/10.12942/lrr-2009-2; open preprint https://arxiv.org/abs/0903.0338.

Analysis

- **Scope.** A long review of 137 pages with 16 figures (stated on the preprint page). Its abstract promises the physics of gravitational waves, how they interact with detectors (bars and interferometers) and how detectors work, the likely sources, the data-analysis methods used to extract signals from noise, and the consequences for physics, astrophysics and cosmology.
- **Structure.** Numbered sections move from an introduction that says what the review is about (Section 1.2), to gravitational-wave observables (Section 2), to data analysis including Bayesian inference (Section 5), to physics with gravitational waves (Section 6), to astrophysics with gravitational waves (Section 7).
- **Time frame.** The abstract says detectors were on an upgrade path expected to give secure detections by 2014, so the review was written before the first detection in 2015. A good analysis notes how this affects what the review can claim, and how a reader should treat predictions in an older review.

Concept map. The map below links "gravitational waves" to the review's main themes.



Summary

Sathyaprakash and Schutz survey the physics of gravitational waves, how bar and interferometer detectors work, the likely sources, the data-analysis methods used to extract signals from noise, and what detections could teach us about physics, astrophysics and cosmology. It is a long, well-illustrated review (137 pages, 16 figures) that works as an entry point to the field.

6. Portfolio Item : Analysis of a scientific poster

Poster analysed. M. Sadun Bordoni, "The extended mass distribution in the Galactic Center," conference poster, Max Planck Institute for Extraterrestrial Physics, Garching (cited on the poster as GRAVITY Collaboration 2024, in prep.). Presented at *Black Holes Inside and Out, The Niels Bohr Institute, August 26 to August 30, 2024.*

Frame element	**Analysis**
Audience and purpose	Written for astronomers and physicists at a conference, to present two results from the GRAVITY collaboration.
Message	No large hidden mass lies inside S2's orbit: the enclosed mass is compatible with zero, with an upper limit of about $1200\,M_\odot$ and there is no evidence for a dark-matter spike.
Structure and reading order	A navy title band, then three horizontal bands, each with a heading, a text panel on the left and figures on the right: (1) precession detected, (2) mass constrained, (3) comparison and outlook. There are no "Aim", "Method" or "Conclusion" headings: the method is reduced to a multi-star fit and two density profiles, which suits specialists but leaves other readers guessing.
Text load	Roughly 400 words of body text plus about 100 words of captions. Paragraphs are full sentences of four to six lines, which is heavy for a poster.
Visuals and colour	Strong: consistent palette; clear headings; the two bar charts share one vertical scale, so the profiles compare at a glance. Weak: the orbit plots mix units (arcseconds in the first, milliarcseconds in the second); star labels near the centre of the first plot are small and crowded; the last graph carries seven curves and a small legend, and its key red marker and label are among the smallest items in the figure.
Language	Precise and consistent in notation, but dense with terms a non-specialist cannot appreciate without help. "Sagittarius A*" and "Sgr A*" are both used. The closing sentence on LISA and EMRI rates assumes the reader already knows both.
Sources and contact	Only one inline citation (Zhang and Amaro-Seoane, 2024) and one source tag "in prep.", with no reference list and no link or QR code to the paper. Contact is clear: an email address, institute name and logos at the foot of the poster.
Overall	**Strength:** a clean, logical three-part layout with well-chosen figures and a result that is easy to find once you scan the page. **Weakness:** the title and the headline number are not combined, and the headline number lacks its qualifications and baseline.

Summary

This is an effective, well-organised specialist poster. Its three-band layout, matched bar charts and clear contact details work well. It falls short in three ways: the title does not state the result, the headline limit is given without its conditions or its predecessor, and some figures and tags are hard to read from a distance. Moving the result into the title would make the poster both clearer and more accurate.

7. Portfolio Item : Analysis of a thesis

Thesis analysed. S. Mitra, *Gravitational waves from inspiraling binaries and cosmological ramifications*, Ph.D. thesis, Savitribai Phule Pune University (At the Inter-University Centre for Astronomy and Astrophysics, IUCAA), 2006. Shodhganga record (open, CC BY-NC-SA 4.0): http://hdl.handle.net/10603/133177.

Introduction

The thesis investigates topics related to gravitational waves generated by inspiralling binary systems and their implications for cosmology. Its keywords include gravitational waves, chirp signals, Chebyshev interpolation, cosmic microwave background (CMB), and CMB anisotropy, indicating the interdisciplinary nature of the research.

The thesis consists of 223 pages and eight numbered chapters, along with an abstract, preliminary sections, an appendix, a list of publications, and a bibliography containing 186 references. Its organisation follows the conventional structure of a doctoral dissertation in theoretical physics and astrophysics.

The following features are particularly noteworthy:

Title and focus: The title clearly identifies the principal research themes and indicates their connection to the broader field of cosmology.

Technical language: Scientific terms such as chirp signal and Chebyshev interpolation communicate specialised concepts precisely. Their effective use requires clear explanations appropriate to the intended readership.

Organisation and structure: The division into chapters, supported by a contents page, lists of figures and tables, an appendix, and references, facilitates navigation and systematic presentation.

Logical presentation: Scientific writing should establish a clear relationship between the research problem, theoretical framework, mathematical methods, results, and conclusions.

Referencing: The extensive bibliography demonstrates the importance of documenting previous research and acknowledging scholarly contributions. References should support the argument rather than merely increase the reference count.

Interdisciplinary approach: The combination of gravitational physics, numerical methods, and cosmology illustrates how different scientific concepts and techniques can be integrated within a research study.

Summary

S. Mitra's doctoral thesis demonstrates the scope and structure of research in gravitational-wave physics and cosmology. Its organisation and technical orientation make it a useful resource for understanding the conventions of scientific writing. This analysis highlights the importance of a clear title, appropriate technical terminology, logical organisation, coherent argumentation, and accurate referencing in preparing an effective academic document.

8. Portfolio Item : Appreciation of a popular science article

Popular article. APS News, "September 2015: Physicists detect gravitational waves for the first time" ("This Month in Physics History"): https://www.aps.org/apsnews/2025/08/detect-gravitational-waves-first-time. For comparison, E. Berti's Physics Viewpoint on the same event, written for physicists rather than the general public, and the 2017 Nobel Prize page.

Appreciation

The article is written for physicists and informed general readers, and it works as a story rather than a summary. Its subject is the first detection of gravitational waves, but it begins with the human drama: the detectors had only just been upgraded and were running when the signal arrived, and the scientists were at first doubtful that it was real. That doubt gives the reader a reason to keep reading. The article then explains why detection is hard by giving a single number, the tiny strain of about 10^{-21}, and relies on a quotation from a named instrument scientist, which makes a technical point feel personal. The detail that the signal, once converted to sound, can be played through a speaker as a rising chirp is the strongest hook: it lets a non-specialist grasp the result through hearing, not equations.

Summary

The APS News article effectively communicates the historic first detection of gravitational waves by combining scientific explanation with human interest. Its narrative structure, accessible language, expert quotations, and the use of the audible chirp signal make a complex scientific discovery understandable and engaging. The article demonstrates how popular science writing can simplify technical concepts without losing their scientific significance.

9. Journals and data sources in gravitational-wave physics

Note This is an optional supporting record for the bibliography database (Unit 3.1)

Introductory Statement: Gravitational-wave physics, the study of ripples in spacetime detected by laser interferometers. Following is a selection of journal titles where gravitational-wave results or reviews are actually published. The data sources listed are those that can be accessed without a subscription.

9.1 Journals

Journal (link)	Publisher	ISSN	Scope relevant here	Open access
Physical Review Letters	APS	0031-9007	Short, high-impact letters; the first detection was published here	Hybrid
Physical Review D	APS	2470-0010	Gravitation, cosmology, particles; longer technical papers	Hybrid
Classical and Quantum Gravity	IOP Publishing	0264-9381	Gravitation, detector instrumentation	Hybrid
The Astrophysical Journal Letters	AAS / IOP	2041-8205	Rapid astrophysics results, including multi-messenger events	Check
The Astrophysical Journal Supplement Series	AAS / IOP	0067-0049	Data releases and catalogues (open-data papers appear here)	Check
Living Reviews in Relativity	Springer	Check on site	Peer-reviewed, regularly updated review articles in relativity	Check
General Relativity and Gravitation	Springer	0001-7701	Theory and applications of general relativity	Hybrid
Monthly Notices of the Royal Astronomical Society	Oxford Univ. Press	0035-8711	Astronomy and astrophysics	Check
Astronomy & Astrophysics	EDP Sciences	0004-6361	Astronomy and astrophysics	Check

Journal (link)	Publisher	ISSN	Scope relevant here	Open access
Pramana – Journal of Physics	Springer, for Indian Academy of Sciences	0304-4289	Indian physics research, including invited reviews	Hybrid

Optional Indian outreach titles: *Resonance* and *Current Science*.

9.2 Data sources

Source (link)	Host	Kind of data	Access / licence note
Gravitational Wave Open Science Center (GWOSC)	LIGO, Virgo, KAGRA collaborations	Calibrated strain time series, event catalogues, data-quality segments	Free; data released under CC BY 4.0
GWOSC tutorials and software	GWOSC	Worked examples for reading and plotting strain data	Free
Gravity Spy data set (Zenodo)	Zenodo (open repository)	Labelled detector-noise glitches for machine learning	Free; check the record for licence
SXS catalogue of simulations	Simulating eXtreme Spacetimes collaboration	Numerical-relativity waveforms of binary black holes	Free; check the site for terms
General Coordinates Network (GCN)	NASA	Rapid alerts and follow-up of transient events	Free
ATNF Pulsar Catalogue	CSIRO (Australia)	Parameters of known pulsars	Free; cite the catalogue paper listed on the site

Summary

Checked each journal's own site for its scope, peer-review statement and ISSN, rather than copying descriptions from third-party lists. Dropped journals that could not be verified, using Think. Check. Submit. as a checklist and the Committee on Publication Ethics standards as a benchmark.

10. How a teacher can assess this portfolio

The portfolio carries 20 of the 75 marks of the semester-end practical evaluation. The other marks are the skill assessment (25), based on the presentation of any one practical from the portfolio, and the viva (30).

Criterion	Strong portfolio	Weak portfolio
Completeness	All seven items, cover, contents, declaration, revision evidence	Missing items or no drafts
Source quality	Primary sources and one review; every link opens; metadata verified against the DOI	Unverified web pages; broken or invented references
Data honesty	Uncertainties stated and explained; no misleading axes; consistent units	Numbers without uncertainty; unlabelled axes
Analysis depth	Specific evidence from the item; own words; strength, weakness and suggestion	Summary only; copied sentences; no judgement
Communication	Different writing for expert and general readers; clean layout	Same register everywhere; cluttered posters
Integrity	Quotations under a sentence, cited; paraphrase changes structure as well as words	Close paraphrase of abstracts or pasted text

11. Sample Viva questions (Critical Thinking and Academic Writing)

The viva carries 30 marks. The following questions assess the student's understanding of the portfolio, scientific reasoning, academic writing, source evaluation, and research communication. A workable pattern is six questions of 5 marks each, drawn from different groups below (suggestion). Students should support their answers with specific examples from their submitted work.

A. Sources and Research Integrity

1. How did you evaluate the reliability of the sources in your bibliography database? Give one example of a source you would reject and justify your decision.
2. Why is a DOI or a repository link not, by itself, sufficient evidence that a scientific claim is correct?
3. What difficulties did you encounter while creating your bibliography database? Demonstrate how you verified and corrected one reference.
4. How would you distinguish plagiarism, close paraphrasing, and legitimate use of a

quotation in your portfolio?

B. Scientific Understanding and Data Interpretation

5. In your data table for GW150914, what do the upper and lower uncertainties in the initial black-hole mass represent? Why should they not automatically be interpreted as standard deviations?
6. The two initial black-hole masses are approximately $36\,M_\odot$ and $29\,M_\odot$, while the final mass is approximately $62\,M_\odot$. How is the difference related to the energy radiated as gravitational waves? Why should the quoted uncertainties also be considered?
7. Why is the gravitational-wave strain of order 10^{-21} so challenging to measure? Explain the basic principle by which an interferometer detects such a signal.
8. Why is matched filtering useful in gravitational-wave detection? How does it differ from simply applying a band-pass filter to the detector data?

C. Critical Analysis of Scientific Writing

9. What distinguishes a research article from a review paper? Illustrate your answer using the two publications analysed in your portfolio.
10. The review paper was published in 2009, before the first direct detection of gravitational waves in 2015. How should a reader interpret its predictions in the light of subsequent discoveries?
11. What distinguishes a doctoral thesis from a journal research article in terms of structure, depth, and documentation? Give examples from the thesis you analysed.
12. In your analysis of the popular science article, how do narrative, expert quotations, and the audible chirp help communicate a complex scientific discovery? What scientific details might be lost through simplification?

D. Visual Communication and Academic Writing Skills

13. Compare your two posters. How do their intended audiences influence the choice of title, text, figures, colours, and technical terminology?
14. Your portfolio includes a deliberately poor graph. Explain how misleading axes, missing units, or inadequate captions can distort the interpretation of scientific data.
15. Choose one paragraph from your portfolio that you revised. What changes did you make to improve clarity, precision, coherence, or conciseness? Justify your revisions.
16. If you had to explain the first detection of gravitational waves to a school student in two minutes, how would your explanation differ from the one you would give at a physics seminar?

E. Reflective Question

17. Which activity in the portfolio most improved your academic writing skills, and what specific evidence demonstrates that improvement?

Guidance for the examiner: Students should answer in their own words and refer to specific entries, figures, references, or drafts in their portfolios. Follow-up questions may be used to probe their reasoning, verify their contribution, and distinguish genuine understanding from memorised answers.

12. All resources used in this sample

Research and review literature

- Abbott *et al.* (2016), first detection: https://doi.org/10.1103/PhysRevLett.116.061102
- Abbott *et al.* (2017), GW170817: https://doi.org/10.1103/PhysRevLett.119.161101
- Aasi *et al.* (2015), Advanced LIGO: https://doi.org/10.1088/0264-9381/32/7/074001
- Abbott *et al.* (2023), open data from the third observing run: https://doi.org/10.3847/1538-4365/acdc9f
- Hulse and Taylor (1975): https://doi.org/10.1086/181708
- Sathyaprakash and Schutz (2009), review: https://doi.org/10.12942/lrr-2009-2 and https://arxiv.org/abs/0903.0338
- Berti (2025), ten years of gravitational-wave astronomy: https://arxiv.org/abs/2509.10395

Popular and commentary sources

- APS News (2025): https://www.aps.org/apsnews/2025/08/detect-gravitational-waves-first-time
- Berti (2016), Physics Viewpoint: https://doi.org/10.1103/Physics.9.17
- Nobel Prize in Physics 2017: https://www.nobelprize.org/prizes/physics/2017/summary/

Thesis and poster

- Mitra (2006), Shodhganga: http://hdl.handle.net/10603/133177
- M. Sadun Bordoni, The extended mass distribution in the Galactic Center, https://share.strong-gr.com/index.php/s/3CnBw9NrRinaKmC

Data and tools

- GWOSC: https://gwosc.org/; tutorials: https://gwosc.org/tutorials/; software: https://gwosc.org/software/
- Gravity Spy data set: https://doi.org/10.5281/zenodo.1486046
- SXS catalogue: https://data.black-holes.org/; NASA GCN: https://gcn.nasa.gov/; ATNF Pulsar Catalogue: https://www.atnf.csiro.au/research/pulsar/psrcat/
- Reference managers: https://www.zotero.org/, https://www.jabref.org/
- Journal-quality checks: https://thinkchecksubmit.org/, https://publicationethics.org/

Acknowledgements

The author used Claude (Anthropic, https://www.anthropic.com/) as a writing assistant during preparation of this portfolio, for drafting text, structuring tables, and improving clarity of language.